

Puducherry Technological University Atal Incubation Centre - PEC Foundation

Supported by Atal Innovation Mission (AIM), NITI Aayog, Government of India, PTU Campus, Puducherry - 605 014.



AIC-PEC FOUNDATION SKILL DEVELOPMENT PROPOSAL

Generative AI and AI Productivity Tools Skill Development Program
(Capacity Building and Technology Enablement Program)

Proposed for
AKT Memorial College of Engineering and Technology

Creating an Instinctive Startup Ecosystem

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ABBREVIATIONS

PTU Puducherry Technological University

AIC-PECF Atal Incubation Centre – Pondicherry Engineering College Foundation

1. ABOUT AIC-PECF

AIC-Pondicherry Engineering College Foundation is a business incubation centre supported by Atal Innovation Mission (AIM) NITI Aayog, Government of India, based out of the scenic coastal city Puducherry.

Puducherry being a Tier-2 city with its economy predominantly dependent on Agriculture, Tourism & MSME sector AIC-PECF is working towards its vision of turning the city and its ecosystem into a deeptech startup hub towards its focus areas: **(Unmanned Aerial Vehicle, Internet of Things, Electronics Design Manufacturing)** complementing the existing industries thereby creating greater employment opportunity improving people life and economic growth.

AIC-Pondicherry Engineering College Foundation supported by AIM – NITI Aayog was officially launched by then Hon’ble Chief Minister of Puducherry Mr. V Narayanasamy in the month of December 2019 during our flagship program Startup Summit Puducherry. Over the past two years the Incubation Centre has worked relentlessly towards reaching its goals by engaging all the ecosystem stake holders from industries, academic institutes, government authorities and investor community in organizing multiple Innovation challenges, entrepreneurship development and training programs for startups and young student entrepreneurs towards pursuing their own ideas and look into entrepreneurship as a career option.

AIC-PECF in addition to its value adding programs also provides comprehensive range of facilities and services such as Technological support, Co-Working space, Rapid Prototyping facility with state-of-the-art additive manufacturing and electronic design and manufacturing machineries and tools to support young innovators in converting their ideas into commercializable product / solution.

Our Vision & Mission

- ✚ Provide world class incubation support to 1st Gen startups that promotes technological innovations to improve people’s life, generate employment and drive sustainable economic growth.
- ✚ To promote and inculcate innovation and entrepreneurship in the UT of Puducherry
- ✚ Leverage Demographic Dividend thereby creating employment and socio-economic growth in Puducherry.

2. PRELUDE

The rapid evolution of emerging technologies is transforming the way industries design products, automate processes, solve complex problems, and deliver innovative solutions. Technologies such as Generative Artificial Intelligence, AI-powered automation, cloud-based development, low-code platforms, collaborative software development, and intelligent digital tools are becoming integral across sectors including manufacturing, healthcare, finance, education, agriculture, retail, and smart infrastructure. As organizations increasingly adopt these technologies, there is a growing demand for graduates who possess practical, industry-relevant skills beyond the conventional engineering curriculum.

While the academic curriculum provides students with a strong foundation in core engineering principles and domain-specific knowledge, several emerging technologies and modern development practices are not extensively covered as part of the prescribed syllabus. This creates a gap between academic learning and the rapidly evolving expectations of industry, where employers seek professionals who can leverage contemporary tools, collaborate effectively, build intelligent solutions, and adapt to evolving technological ecosystems.

Aligned with the vision of the National Education Policy (NEP) 2020, Skill India, Digital India, and the Atmanirbhar Bharat initiative, there is an increasing emphasis on experiential learning, innovation, multidisciplinary education, and industry-driven skill development. Value-added programs play a crucial role in complementing the academic curriculum by exposing students to cutting-edge technologies, practical applications, and hands-on learning experiences that enhance employability and innovation capabilities.

The proposed program aims to bridge this skill gap by providing structured, hands-on learning that equips participants with industry-relevant competencies, enhances technical problem-solving abilities, and prepares them for advanced learning, internships, innovation projects, and emerging career opportunities in embedded systems and smart technologies.

3. PROGRAM OVERVIEW

To implement a structured **Generative AI and AI Productivity Tools Skill Development Program** for the B.Tech Artificial Intelligence & Data Science students of **AKT Memorial College of Engineering and Technology**, aimed at strengthening practical skills in Generative AI, Prompt Engineering, AI-powered productivity tools, Large Language Models (LLMs), AI-assisted application development, and intelligent workflow automation through hands-on, industry-oriented training.

The proposed initiative, "**Generative AI and AI Productivity Tools Skill Development Program**," is a short-term, intensive training program designed and delivered by **AIC – Pondicherry Engineering College Foundation (AIC-PECF)** to bridge the gap between

academic learning and rapidly evolving industry practices in Artificial Intelligence. While students gain a strong theoretical foundation in AI and Data Science through their academic curriculum, emerging technologies such as Large Language Models (LLMs), Prompt Engineering, AI Assistants, AI Agents, Retrieval-Augmented Generation (RAG), and AI-powered productivity platforms are transforming industries and require practical exposure beyond the prescribed syllabus.

The program provides students with hands-on experience using industry-leading AI platforms and tools, enabling them to develop AI-powered solutions, automate routine tasks, enhance productivity, and build practical applications through guided laboratory sessions, case studies, and mini-projects. The training is designed to enhance students' technical competencies, analytical thinking, creativity, and problem-solving abilities while preparing them for internships, research, innovation projects, entrepreneurship, and careers in Artificial Intelligence and emerging digital technologies.

a. Practical Skill Development for AI & DS Students

Students pursuing Artificial Intelligence and Data Science possess strong theoretical knowledge in AI, Machine Learning, Deep Learning, and Data Analytics. However, many have limited opportunities to work with the latest Generative AI technologies and industry-grade AI tools. The proposed program offers practical exposure to Prompt Engineering, AI-powered content generation, AI coding assistants, AI productivity platforms, and workflow automation, enabling students to effectively apply AI concepts in real-world scenarios.

b. Bridging the Gap Between Academic Curriculum and Industry Requirements

Industries across software development, healthcare, finance, manufacturing, education, retail, and business services are rapidly adopting Generative AI to improve efficiency, automate workflows, and accelerate innovation. Organizations increasingly seek graduates who possess practical skills in AI-assisted development, prompt design, AI application integration, and responsible AI usage. The proposed program bridges this gap by introducing students to contemporary AI technologies and industry best practices through structured, hands-on learning experiences.

c. Technical Skill Development & Capacity Building

The program adopts an experiential, project-based learning approach covering:

- Introduction to Generative AI and Large Language Models (LLMs)
- Prompt Engineering Techniques and Best Practices
- AI Productivity Tools (ChatGPT, Gemini, Claude, Perplexity, NotebookLM)
- AI-assisted Coding and Software Development
- AI-powered Content Generation and Research
- AI Workflow Automation and Collaboration Tools
- Mini Project Development using Generative AI

Students will participate in interactive demonstrations, guided hands-on sessions, real-world case studies, collaborative activities, and mini-project development, enabling them to strengthen technical competencies, critical thinking, communication, creativity, and industry readiness.

d. Career Readiness & Institutional Impact

The **Generative AI and AI Productivity Tools Skill Development Program** will enhance the practical capabilities of students at **AKT Memorial College of Engineering and Technology**, preparing them for internships, campus placements, higher education, research, entrepreneurship, and careers in Artificial Intelligence, Data Science, Software Development, Business Analytics, Digital Transformation, and AI-enabled product development. The initiative also supports the institution's objective of producing industry-ready graduates equipped with contemporary technical skills, innovation-oriented thinking, and practical experience in emerging AI technologies.

4.PROGRAM COMPONENTS

I Year: Generative AI & AI Productivity Tools

The proposed **Generative AI & AI Productivity Tools Skill Development Program** comprises the following core components:

1. Introduction to Artificial Intelligence & Generative AI

- Overview of Artificial Intelligence, Machine Learning, and Generative AI.
- Understanding Large Language Models (LLMs) and their real-world applications.
- Introduction to leading AI platforms such as ChatGPT, Gemini, Claude, Perplexity and NotebookLM.

2. Prompt Engineering Fundamentals

- Understanding prompt design principles and best practices.
- Techniques for generating accurate and effective AI responses.
- Practical exercises in content generation, research assistance, and problem-solving.

3. AI Productivity & Digital Collaboration

- AI-assisted documentation, report writing, presentations, and communication.
- AI tools for learning, coding assistance, brainstorming, and project planning.
- Best practices for responsible AI usage and academic integrity.

4. Hands-on AI Applications

- Practical exercises using Generative AI tools for engineering use cases.
- AI-assisted data analysis, content creation, and technical documentation.
- Introduction to AI-powered workflow automation.

5. Mini Project Development

- Development of simple AI-assisted solutions addressing real-world problems.
- Team-based activities emphasizing creativity, collaboration, and innovation.
- Presentation and demonstration of AI-generated solutions.

6. Assessment & Career Awareness

- Practical assessment through AI-based assignments and presentations.
- Evaluation based on creativity, prompt design, and solution quality.
- Career awareness on emerging opportunities in Artificial Intelligence and Digital Technologies.

II Year: AI Agents & Intelligent Application Development

The proposed **AI Agents & Intelligent Application Development Skill Development Program** comprises the following core components:

1. Introduction to AI Agents

- Fundamentals of AI Agents and autonomous AI systems.
- Understanding AI workflows and intelligent task execution.
- Overview of modern AI Agent frameworks.

2. Large Language Models & API Integration

- Working with LLM APIs for intelligent application development.
- Connecting AI services with external applications.
- Practical implementation of AI-powered solutions.

3. Retrieval-Augmented Generation (RAG)

- Introduction to knowledge retrieval and AI-powered search.
- Building intelligent question-answering applications.
- Practical implementation using document-based AI systems.

4. AI Workflow Automation

- Automating repetitive tasks using AI.
- Integration of AI with productivity and collaboration platforms.
- Designing intelligent automation workflows.

5. AI Application Development

- Development of AI-powered applications through guided laboratory sessions.
- Team-based mini projects integrating AI services.
- Testing, validation, and presentation of developed solutions.

6. Assessment & Career Development

- Technical evaluation through project demonstrations.
- Performance assessment based on innovation and implementation.
- Career guidance on AI Development, Intelligent Automation, and AI Product Engineering.

III Year: MLOps & AI Deployment

The proposed **MLOps & AI Deployment Skill Development Program** comprises the following core components:

1. Introduction to MLOps

- Fundamentals of Machine Learning Operations.
- AI model lifecycle management.
- Industry practices in AI deployment.

2. AI Model Deployment

- Deploying AI models using FastAPI and Streamlit.
- Building interactive AI applications.
- API-based AI service deployment.

3. Containerization & Version Control

- Introduction to Docker and containerized applications.
- Git and GitHub for collaborative AI development.
- Managing AI project repositories.

4. AI Monitoring & Optimization

- Monitoring deployed AI models.
- Performance optimization and debugging.

5. Project-Based Learning

- Deployment of end-to-end AI applications.
- Integration of trained models into real-world systems.
- Team-based implementation and validation.

6. Assessment & Career Readiness

- Practical evaluation through deployment projects.
- Technical presentation and demonstration.
- Career guidance on MLOps, AI Engineering, Cloud AI, and DevOps.

IV Year: Enterprise AI, Advanced Applications & Career Readiness

The proposed **Enterprise AI, Advanced Applications & Career Readiness Skill Development Program** comprises the following core components:

1. Enterprise AI Applications

- Understanding the adoption of Artificial Intelligence across industries.
- Exploring enterprise use cases in healthcare, manufacturing, finance, retail, and smart systems.
- Implementing AI-driven solutions for real-world business challenges.

2. AI Application Development & Deployment

- Developing end-to-end AI-powered applications using modern AI tools.
- Integrating Large Language Models (LLMs) into practical applications.
- Introduction to AI model deployment, testing, and performance optimization.

3. AI Tools for Software Engineering

- AI-assisted coding, debugging, documentation, and code optimization.
- Version control, collaborative development, and AI-assisted software lifecycle management.
- Best practices for developing scalable AI-enabled applications.

4. Responsible AI & Industry Best Practices

- Ethical AI, responsible AI principles, data privacy, and AI governance.
- Understanding model limitations, bias, security, and regulatory considerations.
- Best practices for deploying AI solutions in enterprise environments.

5. Capstone Project Development

- Development of an industry-oriented AI application integrating the concepts learned during the program.
- Collaborative implementation, testing, debugging, and solution validation.
- Demonstration of practical problem-solving through real-world AI applications.

6. Technical Assessment & Career Readiness

- Technical evaluation through project demonstrations, presentations, and practical assessment.
- AI-assisted resume building, portfolio enhancement, LinkedIn optimization, and interview preparation.
- Career guidance on opportunities in Artificial Intelligence, Data Science, Machine Learning Engineering, AI Development, MLOps, Cloud AI, Software Engineering, Research, and higher education.

5.PROGRAM STRUCTURE

The proposed **Value-Added Skill Development Programs** are structured as **3-day intensive hands-on training programs**, exclusively designed for the **B.Tech Artificial Intelligence & Data Science students of AKT Memorial College of Engineering and Technology**. Each program is tailored to the academic level of the students, ensuring progressive learning from AI fundamentals to advanced industry applications. Every day combines expert-led technical sessions, practical demonstrations, hands-on exercises, collaborative activities, and mini-project development to enhance technical competency, innovation, and industry readiness.

I Year – Generative AI & AI Productivity Tools

Day	Focus Area	Key Learning & Outcomes
Day 1	Introduction to Generative AI & Prompt Engineering	Introduction to Artificial Intelligence, Generative AI, Large Language Models (LLMs), and Prompt Engineering fundamentals. Students explore AI platforms such as ChatGPT, Gemini, Claude, Perplexity, and NotebookLM while learning effective prompt design techniques.
Day 2	AI Productivity & Engineering Applications	Practical use of AI tools for technical writing, presentations, documentation, coding assistance, research, and data analysis. Students gain hands-on experience in improving productivity using AI-powered tools.
Day 3	Mini Project, Assessment &	Students develop AI-assisted solutions for engineering problems through guided mini-projects, followed by

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	Career Awareness	project presentations, technical evaluation, and an overview of career opportunities in Artificial Intelligence and emerging technologies.
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II Year – AI Agents & Intelligent Application Development

Day	Focus Area	Key Learning & Outcomes
Day 1	Introduction to AI Agents & LLM Applications	Introduction to AI Agents, Large Language Models, AI workflows, and intelligent automation. Students understand how AI agents perform autonomous tasks using modern AI frameworks.
Day 2	RAG & AI Workflow Automation	Hands-on implementation of Retrieval-Augmented Generation (RAG), AI-powered search, workflow automation, and integration of AI services for intelligent application development.
Day 3	AI Application Development & Assessment	Students build simple AI-powered applications through guided practical sessions, followed by project demonstrations, technical evaluation, and discussions on career opportunities in AI Development and Intelligent Automation.

III Year – MLOps & AI Model Deployment

Day	Focus Area	Key Learning & Outcomes
Day 1	Introduction to MLOps & AI Deployment	Fundamentals of Machine Learning Operations (MLOps), AI deployment workflows, version control using Git & GitHub, and deployment strategies for AI applications.
Day 2	Model Deployment & Containerization	Practical sessions on deploying AI applications using Streamlit/FastAPI, Docker fundamentals, application testing, debugging, and deployment best practices.
Day 3	Deployment Project, Assessment & Career Guidance	Students deploy an end-to-end AI application, demonstrate deployment workflows, undergo technical evaluation, and receive guidance on careers in AI Engineering, Cloud AI, and MLOps.

IV Year – AI Innovation & Product Development.

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Day	Focus Area	Key Learning & Outcomes
Day 1	Enterprise AI & Responsible AI	Introduction to enterprise applications of Generative AI across software development, healthcare, finance, manufacturing, and business operations. Students learn Responsible AI practices, AI governance, ethics, data privacy, and secure usage of AI tools in professional environments.
Day 2	AI for Software Development & Workplace Productivity	Hands-on training on AI-assisted coding, debugging, documentation, testing, research, report generation, and workflow automation using leading AI tools. Students gain practical exposure to improving productivity through AI-enabled development practices and collaborative tools.
Day 3	Capstone Implementation, Technical Assessment & Career Readiness	Students develop and demonstrate an AI-assisted solution addressing a real-world engineering problem. The program concludes with technical evaluation, project demonstration, resume optimization using AI, interview preparation, industry best practices, and guidance on career opportunities in Artificial Intelligence, Data Science, Software Engineering, AI Operations, and Digital Transformation.

6. PROGRAM EXECUTION MODALITY

1. Program Planning & Coordination

AIC – Pondicherry Engineering College Foundation (AIC-PECF), in coordination with **AKT Memorial College of Engineering and Technology**, shall jointly finalize the training schedule, participant nominations, laboratory requirements, software access, and logistical arrangements to ensure the effective execution of the Value Added Skill Development Programs for the B.Tech Artificial Intelligence & Data Science students across all four academic years.

2. Training Implementation

The programs shall be delivered by experienced faculty members, industry experts, and technical trainers from AIC-PECF through an outcome-based learning methodology comprising instructor-led technical sessions, live demonstrations, hands-on practical exercises, guided laboratory activities, case studies, collaborative learning, and mini-project development. The training shall emphasize experiential learning using

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contemporary Artificial Intelligence tools and platforms to strengthen students' practical understanding of emerging AI technologies and their real-world applications.

3. Program Duration & Delivery Schedule

Each Value-Added Course shall be conducted as a **three-day intensive skill development program**, consisting of structured technical sessions, practical hands-on activities, guided exercises, mini-projects, and technical assessments. Separate training modules shall be delivered for First, Second, Third, and Fourth-Year students, with content tailored to their academic level and aligned with the institution's academic calendar to ensure maximum participation and effective learning outcomes.

4. Laboratory Infrastructure & Learning Resources

AIC – Pondicherry Engineering College Foundation (AIC-PECF) shall provide the required training content, AI software platforms, learning resources, case studies, practical exercises, datasets, and training manuals necessary for the successful execution of the Value Added Skill Development Programs. **All participating students shall bring their own laptops**, pre-configured with the software and tools specified by AIC-PECF prior to the commencement of the training, to facilitate uninterrupted hands-on learning and practical exercises. AKT Memorial College of Engineering and Technology shall provide the necessary training venue, high-speed internet connectivity, multimedia facilities, projection systems, power supply, and other essential infrastructure required for the smooth conduct of the program.

5. Assessment, Certification & Program Management

AIC-PECF shall serve as the nodal implementation agency responsible for program planning, coordination, training delivery, monitoring, assessment, and documentation. Participant learning shall be evaluated through practical assignments, hands-on exercises, mini-projects, technical assessments, and final demonstrations. Students who successfully complete the program and satisfy the prescribed assessment criteria shall be awarded a **Certificate of Participation/Completion**, jointly issued by AIC – Pondicherry Engineering College Foundation (AIC-PECF) and AKT Memorial College of Engineering and Technology, recognizing the successful completion of the respective Value Added Skill Development Program

7. PROGRAM FINANCIALS

Proposed Fund Utilization Period: 3 Days (Training Implementation)

Sl No	Particulars	Amount (INR)
1	Value Added Skill Development Program (Inclusive of Expert Training, Learning Resources, AI Software Platforms, Course Materials, Assessment, Certification, and Program Management)	₹ 850 Per Student (Inclusive of GST)

The proposed program fee includes expert-led technical sessions, hands-on practical training, access to AI tools and software platforms (where applicable), digital learning resources, course materials, technical assessments, certification, logistics for program execution, and overall program coordination and management by AIC-Pondicherry Engineering College Foundation (AIC-PECF). **Accommodation and food for the resource persons will be arranged by AKT Memorial College of Engineering and Technology.**

7. IMPACT & OUTCOME MEASUREMENT

The proposed **Value Added Skill Development Programs** are designed to strengthen the technical, analytical, and industry-oriented competencies of B.Tech Artificial Intelligence & Data Science students by providing hands-on exposure to emerging Artificial Intelligence technologies, Generative AI tools, AI-assisted application development, intelligent automation, and contemporary software development practices. The programs aim to bridge the gap between academic learning and industry expectations by enabling students to apply AI concepts to real-world scenarios, improve problem-solving capabilities, and enhance their readiness for internships & placements.

Expected Outcomes

- ✓ Enhanced practical knowledge of Generative AI, Large Language Models (LLMs), Prompt Engineering, AI productivity tools, and emerging AI technologies.
- ✓ Improved competency in developing AI-assisted solutions, automating workflows, integrating AI tools, and applying AI concepts to solve real-world engineering problems.
- ✓ Practical exposure to modern AI development practices, intelligent applications, AI-powered software development, and responsible AI through hands-on laboratory sessions and guided exercises.
- ✓ Strengthened analytical thinking, problem-solving abilities, critical reasoning, and collaborative learning through application-oriented mini-projects and technical activities.
- ✓ Development of practical AI-based mini-projects, digital portfolios, technical documentation, and project deliverables demonstrating industry-relevant skills and learning outcomes.

Impact Metrics:

Impact Segment	Key Measurement Areas
Student Skill Development	Students demonstrate hands-on competency in Generative AI, Prompt Engineering, AI productivity tools, and AI-assisted application development.
Practical Application	Students successfully complete hands-on exercises, AI-based assignments, workflow automation tasks, and mini-projects.
Technical Confidence	Students demonstrate improved ability to apply AI tools, solve real-world problems, and develop AI-assisted solutions independently.
Digital Literacy	Students gain proficiency in AI platforms, digital collaboration tools, responsible AI practices, and technical documentation.
Portfolio Creation	Students compile AI-generated outputs, practical assignments, mini-projects, and technical documentation into individual portfolios.
Experiential Learning	Students engage in hands-on, application-oriented learning, reinforcing theoretical concepts through practical AI-based problem-solving.

8. PREVIOUS PROGRAM OUTCOMES

- ✓ Enhanced AIC – Pondicherry Engineering College Foundation (AIC-PECF) has successfully organized and delivered several industry-oriented skill development and awareness programs, resulting in the following outcomes:
- ✓ Successfully conducted **Embedded Systems Skill Development Programs** with hands-on learning and practical implementation.
- ✓ Delivered **Drone Technology Training Programs** to provide practical exposure to UAV technologies and applications.
- ✓ Organized **Artificial Intelligence Skill Development Programs** focusing on emerging AI tools and industry-relevant technologies.
- ✓ Conducted **Entrepreneurship Awareness Programs** to promote innovation, startup awareness, and entrepreneurial thinking among students.

9. CAREER PATHWAYS BEYOND TRAINING

Upon successful completion of the **Value Added Skill Development Programs**, students will have opportunities to further enhance their technical competencies and pursue advanced learning and career opportunities in Artificial Intelligence and emerging technologies.

✓ **Advanced AI Learning**

Students demonstrating strong interest may pursue advanced learning in Generative AI, Machine Learning, AI Agents, MLOps, Data Science, Cloud AI, and other emerging technologies through specialized training and certification programs.

✓ **Project Development & Practical Applications**

Students will be encouraged to develop AI-enabled applications, mini-projects, and practical solutions, strengthening their skills in AI implementation, application development, automation, and problem-solving.

✓ **Internship & Industry Exposure**

Students may be connected with AIC-Pondicherry Engineering College Foundation (AIC-PECF)'s startup ecosystem and industry network for internships, live projects, technical mentoring, and practical exposure in Artificial Intelligence, Data Science, Software Development, and Digital Technologies.

✓ **Career & Higher Education Opportunities**

The program provides a strong foundation for pursuing higher education, professional certifications, research opportunities, internships, and careers in Artificial Intelligence, Machine Learning, Data Science, AI Engineering, Software Development, Digital Transformation, and other emerging technology domains.

10. CONCLUSION

By complementing the academic curriculum with industry-oriented learning, practical exercises, and application-based mini-projects, the programs will strengthen students' technical proficiency, analytical thinking, problem-solving abilities, and digital competencies. The initiative aims to enhance students' industry readiness, preparing them for internships, higher education, research opportunities, and successful careers in Artificial Intelligence, Data Science, Software Development, and other emerging technology domains.

ANNEXURE -1

Detailed Curriculum

Year	Day	Module	Description	Hands-on Training	Outcome
I Year	Day 1	Introduction to Generative AI & Prompt Engineering	Fundamentals of AI, Generative AI, LLMs, and Prompt Engineering.	Hands-on with ChatGPT, Gemini, Claude, Perplexity, and NotebookLM.	Understand AI fundamentals and effective prompt engineering techniques.
	Day 2	AI Productivity & Engineering Applications	AI-assisted documentation, coding, research, and productivity tools.	AI-based report writing, presentations, documentation, and coding assistance.	Develop practical skills in applying AI tools for academic and professional tasks.
	Day 3	AI Mini Project & Assessment	AI-assisted problem-solving and project development.	Development and presentation of an AI-enabled mini project.	Demonstrate practical competency in Generative AI applications.
II Year	Day 1	AI Agents & Intelligent Workflows	Fundamentals of AI Agents, LLMs, APIs, and intelligent workflows.	Creating AI assistants using modern AI platforms.	Understand AI Agent concepts and intelligent automation.
	Day 2	RAG & AI Automation	Retrieval-Augmented Generation (RAG) and AI workflow automation.	Building document-based AI assistants and workflow automation exercises.	Gain practical experience in AI-powered knowledge retrieval and automation.
	Day 3	AI Application Development	Development of AI-assisted applications and intelligent solutions.	Mini project integrating AI tools into practical applications.	Develop AI-enabled applications using industry-relevant technologies.

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Year	Day	Module	Description	Hands-on Training	Outcome
III Year	Day 1	Introduction to MLOps	AI deployment lifecycle, version control, and collaborative development.	Git & GitHub for AI projects.	Understand AI deployment workflows and version control practices.
	Day 2	AI Deployment & Containerization	AI deployment using Streamlit, FastAPI, and Docker.	Deploying AI applications using modern deployment tools.	Gain practical skills in AI application deployment.
	Day 3	Deployment Project & Assessment	End-to-end AI deployment and evaluation.	Capstone deployment project and technical assessment.	Demonstrate competency in MLOps and AI deployment practices.
IV Year	Day 1	Enterprise AI & Responsible AI	Enterprise AI applications, Responsible AI, ethics, governance, and data privacy.	Enterprise AI case studies and practical exercises.	Understand enterprise AI adoption and responsible AI practices.
	Day 2	AI-Assisted Software Development	AI-assisted coding, debugging, documentation, testing, and productivity.	Hands-on with AI coding assistants and software development tools.	Develop practical skills in AI-enabled software engineering.
	Day 3	Capstone Project & Career Readiness	AI solution development, technical assessment, resume enhancement, and interview preparation.	AI-based capstone project, technical presentation, resume optimization, and mock interview.	Demonstrate industry-ready AI competencies and career preparedness.

Training Duration

- **Duration:** 3 Days
- **Training Hours:** 8 Hours per Day (24 Hours Total)